

PayForMe — MTN MoMo Integration

Version: 1.0

Provider: MTN Mobile Money (Collections API)

Prerequisite: Complete [Company onboarding requirements](#) and MTN merchant / API approval

1. Overview

MoMo powers:

- **Wallet top-ups** (user deposits)
- **Subscription payments**
- USSD / app approval flow on the customer's phone

Configure with `PAYMENT_PROVIDER=momo` . For local dev without keys, use `PAYMENT_PROVIDER=log` .

2. Prerequisites

Requirement	Details
MTN MoMo Developer account	https://momodeveloper.mtn.com
Collections product subscription	Subscription key from developer portal
API User + API Key	Created in MTN portal
HTTPS callback URL	Public webhook for payment status
Company KYB	Incorporation cert, TIN, business license — see company onboarding .
Verified MoMo test number	Sandbox uses simulated approval (no real USSD to phone in sandbox)

3. Environment variables

```
PAYMENT_PROVIDER=momo

MOMO_SUBSCRIPTION_KEY=your-subscription-key
MOMO_API_USER=your-api-user-uuid
MOMO_API_KEY=your-api-key

# Sandbox (development)
MOMO_API_URL=https://sandbox.momodeveloper.mtn.com
MOMO_TARGET_ENVIRONMENT=sandbox
MOMO_CURRENCY=EUR

# Production (Ghana) – example
# MOMO_API_URL=https://proxy.momoapi.mtn.com
# MOMO_TARGET_ENVIRONMENT=mtnghana
```

```
# MOMO_CURRENCY=GHS
```

```
MOMO_CALLBACK_URL=https://yourdomain.com/api/webhooks/payments/momo
```

Variable	Sandbox	Production
MOMO_API_URL	https://sandbox.momodeveloper.mtn.com	MTN production proxy URL
MOMO_TARGET_ENVIRONMENT	sandbox	mtnghana (or per MTN docs)
MOMO_CURRENCY	EUR	GHS
MOMO_CALLBACK_URL	ngrok or public dev URL	HTTPS production domain

4. Payment flow

```
User → Wallet → Deposit (MoMo account selected)
→ POST /api/payments/deposit
→ momo-payment.service / momo.service
→ POST {MOMO_API_URL}/collection/v1_0/requesttopay
→ User approves on phone (production) or sandbox simulator
→ POST /api/webhooks/payments/momo
→ completeWalletDeposit() → wallet credited + notification
```

Status polling: GET /api/payments/deposit?reference=...

5. API endpoints (MTN)

MTN endpoint	Purpose
POST /collection/token/	OAuth access token
POST /collection/v1_0/requesttopay	Initiate collection
GET /collection/v1_0/requesttopay/{referenceId}	Check status

PayForMe webhook:

Endpoint	Auth
POST /api/webhooks/payments/momo	HMAC x-signature with MOMO_API_KEY

6. Sandbox vs production

	Sandbox	Production
URL	sandbox.momodeveloper.mtn.com	MTN production API host
Currency	EUR	GHS

Real USSD prompt	No — use developer portal simulator	Yes
Callback	Must be publicly reachable (ngrok for local)	HTTPS required

Important: Sandbox does **not** send real payment prompts to customer phones. Use MTN's sandbox UI to approve/decline test payments.

7. User experience

1. User adds **verified MoMo account** in Settings
2. User enters deposit amount in Wallet
3. App shows: *"Approve the prompt on your phone"*
4. On success: in-app + email notification, balance updates
5. **Bank account deposits** use Partner Bank API — not MoMo (see [bank-partner-api.md](#))

8. Webhook payload (simplified)

```
{
  "externalId": "MOMO-ABC12345",
  "amount": 100,
  "currency": "GHS",
  "status": "SUCCESSFUL",
  "payer": {
    "partyIdType": "MSISDN",
    "partyId": "233241234567"
  },
  "timestamp": "2026-07-25T12:00:00.000Z"
}
```

Idempotency: `externalId` maps to `WalletTransaction.reference`.

9. Troubleshooting

Issue	Fix
"MoMo not configured"	Set all three: MOMO_SUBSCRIPTION_KEY, MOMO_API_USER, MOMO_API_KEY
No prompt on phone	Expected in sandbox — use simulator
Webhook not received	Check MOMO_CALLBACK_URL is public HTTPS; verify signature
Wrong API host	Use <code>sandbox.momodeveloper.mtn.com</code> not old <code>sandbox.momoapi.mtn.com</code>
Deposit stuck PENDING	Poll status endpoint; check webhook logs
Bank deposit fails	Bank accounts cannot use MoMo — use bank deposit instructions

10. Security

- Store API keys only in server environment variables

- Verify webhook HMAC signature
 - Use unique `externalId` per transaction
 - Rotate keys if compromised
-

11. Related files

Path	Purpose
<code>lib/services/payment/momo.service.ts</code>	MTN API client
<code>lib/services/payment/momo-payment.service.ts</code>	Wallet deposit orchestration
<code>app/api/webhooks/payments/momo/route.ts</code>	Payment webhook
<code>app/api/payments/deposit/route.ts</code>	User deposit API

12. Related documentation

- [Company onboarding](#)
- [Bank Partner API](#) (bank deposits/withdrawals)
- [Platform integrations guide](#) (combined PDF)